

一、有cp的job重启

代码如下：

```

    @Override
    public void open(Configuration parameters) throws Exception {
        super.open(parameters);
        this.dataKeyConverter.prepare(getRuntimeContext());
        this.dataIndexConverter.prepare(getRuntimeContext());
        this.validShowCounter = getRuntimeContext().getMetricGroup().counter( name: "parserValidShowCounter");
        this.invalidShowCounter = getRuntimeContext().getMetricGroup().counter( name: "parserInvalidShowCounter");
        ClassLoader fatherLoader = this.getClass().getClassLoader();
        LOGGER.info("当前jobid: " + getRuntimeContext().getJobId() + ",当前重试attempNUM: " + getRuntimeContext().getAttemptNumber());
        LOGGER.info("当前类的名称: " + fatherLoader.getClass().getName() + ";" + fatherLoader.toString());
        LOGGER.info("当前类父加载器名称: " + fatherLoader.getParent().getClass().getName() + ";" + fatherLoader.getParent().toString());
        LOGGER.info("当前类爷加载器名称: " + fatherLoader.getParent().getParent().getClass().getName() + ";" + fatherLoader.getParent().getParent().toString());
    }

    @Override
    public void flatMap(Tuple4<UnitedEvent, String, Integer, Long> value, Collector<Tuple3<DataKey, DataIndex, String>> out) {
        if (k++ == 30000 && getRuntimeContext().getIndexOffsetSubtask()==1) {
            throw new RuntimeException("this must be dead!!");
        }
        final UnitedEvent unitedEvent = value.f0;
        final String topic = value.f1;
    }

```

第一次启动

```

[] - 当前jobid: c3be2844c2b3bd740a86b98e0f27e3c7,当前重试attempNUM: 0
[] - 当前类的名称: org.apache.flink.util.ChildFirstClassLoader;org.apache.flink.util.ChildFirstClassLoader
[] - 当前类父加载器名称: sun.misc.Launcher$AppClassLoader;sun.misc.Launcher$AppClassLoader@2503dbd1
[] - 当前类爷加载器名称: sun.misc.Launcher$ExtClassLoader;sun.misc.Launcher$ExtClassLoader@6442b0af

```

失败后第二次启动

```

- 当前jobid: c3be2844c2b3bd740a86b98e0f27e3c7,当前重试attempNUM: 1
- 当前类的名称: org.apache.flink.util.ChildFirstClassLoader;org.apache.flink.util.ChildFirstClassLoader
- 当前类父加载器名称: sun.misc.Launcher$AppClassLoader;sun.misc.Launcher$AppClassLoader@2503dbd1
- 当前类爷加载器名称: sun.misc.Launcher$ExtClassLoader;sun.misc.Launcher$ExtClassLoader@6442b0af

```

第六次重启

```

- 当前jobid: c3be2844c2b3bd740a86b98e0f27e3c7,当前重试attempNUM: 6
- 当前类的名称: org.apache.flink.util.ChildFirstClassLoader;org.apache.flink.util.ChildFirstClassLoader
- 当前类父加载器名称: sun.misc.Launcher$AppClassLoader;sun.misc.Launcher$AppClassLoader@2503dbd1
- 当前类爷加载器名称: sun.misc.Launcher$ExtClassLoader;sun.misc.Launcher$ExtClassLoader@6442b0af

```

cancel掉重启

```

2023-05-23 11:24:42,971 [flink-akka.actor.default-dispatcher-15] INFO org.apache.flink.runtime.taskexecutor.slot.TaskSlotTableImpl [] - Free slot TaskSlot(index:2, state:ACTIVE, resource profile: 1.498gb (1608599464 bytes), taskOffHeapMemory=0 bytes, managedMemory=64.000mb (67108864 bytes), networkMemory=181.120mb (189918087 bytes)), allocationId: ea966eb20d54bcadefdd2e7a
2023-05-23 11:24:42,979 [flink-akka.actor.default-dispatcher-15] INFO org.apache.flink.runtime.taskexecutor.slot.TaskSlotTableImpl [] - Free slot TaskSlot(index:4, state:ACTIVE, resource profile: 1.498gb (1608599464 bytes), taskOffHeapMemory=0 bytes, managedMemory=64.000mb (67108864 bytes), networkMemory=181.120mb (189918087 bytes)), allocationId: c6dd8c93f097ebdeff80297
2023-05-23 11:24:42,980 [flink-akka.actor.default-dispatcher-15] INFO org.apache.flink.runtime.taskexecutor.slot.TaskSlotTableImpl [] - Free slot TaskSlot(index:0, state:ACTIVE, resource profile: 1.498gb (1608599464 bytes), taskOffHeapMemory=0 bytes, managedMemory=64.000mb (67108864 bytes), networkMemory=181.120mb (189918087 bytes)), allocationId: c83e859d41cba1abeeab3ac
2023-05-23 11:24:42,981 [flink-akka.actor.default-dispatcher-15] INFO org.apache.flink.runtime.taskexecutor.slot.TaskSlotTableImpl [] - Free slot TaskSlot(index:3, state:ACTIVE, resource profile: 1.498gb (1608599464 bytes), taskOffHeapMemory=0 bytes, managedMemory=64.000mb (67108864 bytes), networkMemory=181.120mb (189918087 bytes)), allocationId: a37ea4fa29e9b258a32640ae
2023-05-23 11:24:42,982 [flink-akka.actor.default-dispatcher-15] INFO org.apache.flink.runtime.taskexecutor.slot.TaskSlotTableImpl [] - Free slot TaskSlot(index:6, state:ACTIVE, resource profile: 1.498gb (1608599464 bytes), taskOffHeapMemory=0 bytes, managedMemory=64.000mb (67108864 bytes), networkMemory=181.120mb (189918087 bytes)), allocationId: bd9701140adf84d2627364df
2023-05-23 11:24:42,982 [flink-akka.actor.default-dispatcher-15] INFO org.apache.flink.runtime.taskexecutor.slot.TaskSlotTableImpl [] - Free slot TaskSlot(index:7, state:ACTIVE, resource profile: 1.498gb (1608599464 bytes), taskOffHeapMemory=0 bytes, managedMemory=64.000mb (67108864 bytes), networkMemory=181.120mb (189918087 bytes)), allocationId: 0a3f03df8a2aa59bce4c6888
2023-05-23 11:24:42,983 [flink-akka.actor.default-dispatcher-15] INFO org.apache.flink.runtime.taskexecutor.slot.TaskSlotTableImpl [] - Free slot TaskSlot(index:1, state:ACTIVE, resource profile: 1.498gb (1608599464 bytes), taskOffHeapMemory=0 bytes, managedMemory=64.000mb (67108864 bytes), networkMemory=181.120mb (189918087 bytes)), allocationId: d12a78fa76207d6fb681694e
2023-05-23 11:24:42,984 [flink-akka.actor.default-dispatcher-15] INFO org.apache.flink.runtime.taskexecutor.slot.TaskSlotTableImpl [] - Free slot TaskSlot(index:5, state:ACTIVE, resource profile: 1.498gb (1608599464 bytes), taskOffHeapMemory=0 bytes, managedMemory=64.000mb (67108864 bytes), networkMemory=181.120mb (189918087 bytes)), allocationId: 77b11a87a7eab2d49541b6fe
2023-05-23 11:24:42,984 [flink-akka.actor.default-dispatcher-15] INFO org.apache.flink.runtime.taskexecutor.DefaultJobLeaderService [] - Remove job c3be2844c2b3bd740a86b98e0f27e3c7 from job leader table
2023-05-23 11:24:42,984 [flink-akka.actor.default-dispatcher-15] INFO org.apache.flink.runtime.leaderretrieval.DefaultLeaderRetrievalService [] - Stopping DefaultLeaderRetrievalService

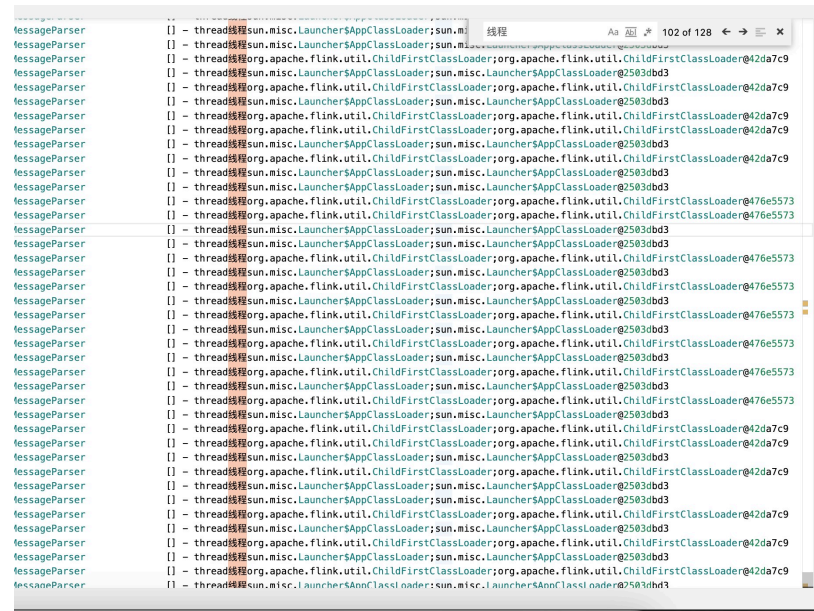
```

– 当前jobid: 5603313f595f02abe7c9afb42c8a0b62, 当前重试attempNUM: 0

– 当前类爷加载器名称: sun.misc.Launcher\$ExtClassLoader; sun.misc.Launcher\$ExtClassLoader@6442b0af

可以发现job本身出错重启并不会改变类加载器，只有手动cancel掉重启child才会重新加载。

二、无cp的job重启



可以看到，childfirst的classloader因为job发生了重启，classloader对象发生了改变，说明只要job重启，那么childfirst的classloader便会发生回收。而appClassLoader作为父类加载器，并没有被回收，说明它的生命周期是与JVM一样，job关闭后还会保留一段时间。

```
105 @Documentation.Section(Documentation.Sections.EXPERT_CLASS_LOADING)
106 public static final ConfigurationList<String> ALWAYS_PARENT_FIRST_LOADER_PATTERNS =
107     ConfigurationKey("classloader.parent-first-patterns.default")
108         .stringType()
109         .toList()
110         .defaultValues(
111             ArrayUtils.concat(
112                 new String[] {
113                     "java.",
114                     "scala.",
115                     "org.apache.flink.",
116                     "com.esotericsoftware.kryo",
117                     "org.apache.hadoop.",
118                     "javax.annotation.",
119                     "org.xml.",
120                     "javax.xml.",
121                     "org.apache.xerces.",
122                     "org.w3c.",
123                     "org.rocksdb."
124                 },
125                 PARENT_FIRST_LOADING_PATTERNS
126             )
127         ).withDeprecatedKeys("classloader.parent-first-patterns")
128         .withDescription(
129             "A (semicolon-separated) list of patterns that specifies which classes should always be"
130             + " resolved through the parent ClassLoader first. A pattern is a simple prefix that is checked against"
131             + " the fully qualified class name. This setting should generally not be modified. To add another pattern we"
132             + " recommend to use \\\"classloader.parent-first-patterns.additional\\\" instead.");
133 @Documentation.Section(Documentation.Sections.EXPERT_CLASS_LOADING)
```

无标签